

Accurate Semi-supervised Automatic Speech Recognition via Multi-hypotheses-based Curriculum Learning

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Abstract. How can we accurately transcribe speech signals into texts when only a portion of them are annotated? ASR (Automatic Speech Recognition) systems are extensively utilized in many real-world applications including automatic translation systems and transcription services. Due to the exponential growth of available speech data without annotations and the significant costs of manual labeling, semi-supervised ASR approaches have garnered attention. Such scenarios include transcribing videos in streaming platforms, where a vast amount of content is uploaded daily but only a fraction of them are transcribed manually. Previous approaches for semi-supervised ASR use a pseudo labeling scheme to incorporate unlabeled examples during training. Nevertheless, their effectiveness is restricted as they do not take into account the uncertainty linked to the pseudo labels when using them as labels for unlabeled cases. In this paper, we propose MOCA (MULTI-HYPOTHESES-BASED CURRICULUM LEARNING FOR SEMI-SUPERVISED ASR), an accurate framework for semi-supervised ASR. MOCA generates multiple hypotheses for each speech instance to consider the uncertainty of the pseudo label. Furthermore, MOCA considers the various degrees of uncertainty in pseudo labels across speech instances, enabling a robust training on the uncertain dataset. Extensive experiments on real-world speech datasets show that MOCA successfully improves the transcription performance of previous ASR models.

1 Introduction

Given speech signals, how can we train an accurate transcription model in a semi-supervised setting? Speech is one of the most fundamental and prevalent forms of human communication. To efficiently utilize the extensive amount of speech data, the development of automatic speech recognition (ASR) methods is essential. ASR models are applied to various fields, including voice search [20], speech command recognition [4], automatic transcription of spoken content [12], information extraction [24], and automatic translation [6].

Recently, semi-supervised ASR is actively studied due to the excessive costs associated with manual labeling and the exploding amount of unlabeled speeches

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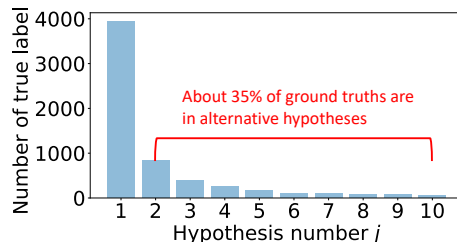


Fig. 1. Case study on LJSpeech dataset showing the distribution of true labels in the 10-best hypotheses. Note that the ground-truth labels appear frequently in alternative hypotheses, rather than being limited to the 1-best hypothesis.

[22]. Such scenarios often arise in customer service calls, where numerous interactions occur daily, yet only specific calls are manually transcribed for quality assurance. Semi-supervised ASR aims to leverage unlabeled data in addition to a few labeled data for training. Semi-supervised ASR improves existing ASR approaches by removing the requirement for fully observed labels, thus expanding the range of applications to more challenging scenarios.

Previous approaches for semi-supervised ASR generate pseudo labels for the unlabeled instances with an initial ASR model that has been trained using a few labeled speech instances [15]. The pseudo labels are then used to further retrain the ASR model [23]. The main limitation of those approaches is that they consider only the 1-best ASR hypothesis, ignoring the power of alternative hypotheses with potentially fewer errors. To further elaborate this idea, we show the distribution of true labels in Fig 1 where the x-axis represents the j^{th} -best hypothesis of 10-best hypotheses, and y-axis represents the number of true labels found in each hypothesis. Note that about 35% of ground-truths are overlooked with 1-best-based approaches in LJSpeech. To prevent such information loss, there arises the need for approaches incorporating multiple hypotheses.

In this work, we propose MOCA (MULTI-HYPOTHESES-BASED CURRICULUM LEARNING FOR SEMI-SUPERVISED ASR), an accurate method for semi-supervised ASR. MOCA incorporates multi-hypotheses-based pseudo labels for each unlabeled instance to reflect uncertainties inherently present in the 1-best pseudo label. MOCA reduces the heavy reliance on the quality of the generated pseudo labels by curriculum learning with our novel difficulty score for each speech instance. Since easier instances have lower uncertainty, MOCA initially enhances the accuracy of the ASR model with relatively certain examples and then addresses uncertain ones. This enables the ASR model to acquire more information before learning uncertain and difficult examples, allowing robust training. That is, the model is less affected by the quality of pseudo labels for difficult instances. Our contributions are summarized as follows:

- **Method.** MOCA effectively considers the uncertainties of the seed model by multi-hypothesis-based pseudo labels. MOCA further reduces the heavy reliance on the quality of pseudo labels by curriculum learning, proposing a novel hardness score for each speech instance.

- **Theory.** We conduct a theoretical analysis of the loss function of MOCA by comparing it with a traditional ASR loss.
- **Experiments.** We perform various experiments and show that MOCA successfully improves the transcription performance of previous ASR models.

The code and datasets are available at <https://github.com/asrbest2023/MOCA>.

2 Related Works

2.1 Automatic Speech Recognition Methods

Advancements in speech recognition have been invigorated by the emergence of unsupervised pre-training methods that extract general features from speeches [1, 9, 17]. The models are trained using large amounts of unlabeled audio data, and the representations are subsequently used in an end-to-end manner to enhance the performance of many downstream tasks including speech emotion recognition [11], disease detection [8], voice conversion [14], etc. The representations based on the inherent properties of the audio are then used to yield probable transcriptions, which are called hypotheses in ASR [10].

Previous methods for semi-supervised ASR employ the 1-best hypothesis approach. Higuchi et al. [7] utilize a pair of online and offline models such that they learn better representations for ASR from interactions. Park et al. [16] adopt Noisy Student Training for ASR with various levels of augmentation on input features. However, relying on a single model prediction does not capture the potential diversity of choices that the model can generate through predictions.

In ASR, wav2vec [19] and wav2vec 2.0 [3] are widely used as pre-trained general-feature extraction models due to their high performance [2, 18]. The model is trained in a contrastive learning manner, distinguishing similar and dissimilar audio pairs. The wav2vec and wav2vec 2.0 models take a speech signal $x_i = [x_{i,1}, \dots, x_{i,T}]$ where $x_{i,j} \in \mathbb{R}$ is a sampled value of the speech signal at time $j \in \{1, 2, \dots, T\}$ as their input. The encoder of them produces feature representation $h_i = [h_{i,1}, \dots, h_{i,T}]$ for each x_i , which is then mapped into a list of states $y_i = [y_{i,1}, \dots, y_{i,T}]$ by a decoder. Any decoder such as MLP can be used. The j -th element $y_{i,j} \in \mathbb{R}^C$ of y_i is the prediction vector at timestamp j of a speech x_i where C is the number of alphabets and other characters such as a blank, question mark, etc.

2.2 Connectionist Temporal Classification (CTC) Loss

To measure the difference between the predicted list of states $y_i = [y_{i,t} \mid t = 1, \dots, T]$ and the ground-truth text label l_i for each speech instance x_i , the connectionist temporal classification (CTC) loss have been widely used in ASR tasks [21]. The loss enables the model to learn the alignment between input audio frames and output characters without pre-segmented data.

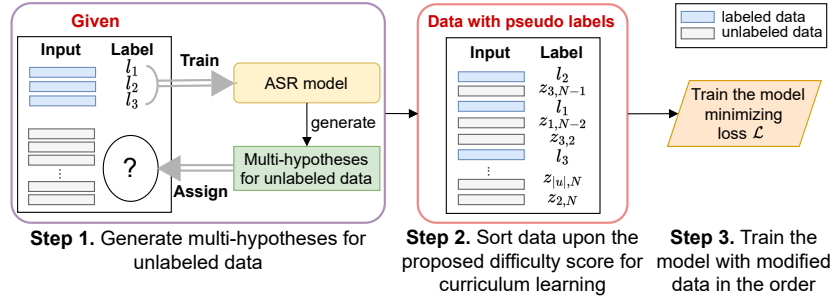


Fig. 2. Overall structure of MOCA. MOCA first trains an ASR model with a few speech examples with labels. Then MOCA utilizes the trained ASR model to generate multiple hypotheses for each unlabeled instance. MOCA retrain the ASR model utilizing the labeled and unlabeled instances with multiple hypotheses minimizing the loss function $\mathcal{L}$. Curriculum learning strategy with our new difficulty score is used during training to relieve the heavy reliance on the pseudo labels.

The CTC algorithm aims to maximize the likelihood of generating a ground-truth transcription l_i when presented with a speech instance x_i as follows:

$$p(l_i | x_i) = \sum_{\pi_i \in \mathcal{B}^{-1}(l_i)} p(\pi_i | x_i) \quad (1)$$

where $\mathcal{B}$ is a function that merges y_i into a transcription by eliminating blanks and duplicated labels. Thus, $\mathcal{B}^{-1}(l_i)$ represents a set of all possible sequences of prediction vectors that merges to the label l_i . Then the CTC loss, which is the negative log likelihood of $p(l | x)$, is expressed as follows:

$$\mathcal{L}_{\text{CTC}} = - \sum_{i=1}^M \log p(l_i | x_i) \quad (2)$$

where M is the number of data instances.

3 Proposed Method

We propose MOCA (MULTI-HYPOTHESES-BASED CURRICULUM LEARNING FOR SEMI-SUPERVISED ASR), an accurate method for semi-supervised ASR. We illustrate the overall process of MOCA in Figure 2. MOCA first trains an ASR model with the labeled speech instances, which is used for generating pseudo labels for unlabeled ones. Then MOCA retrain the ASR model with both labeled and unlabeled instances, upon the defined order considering the difficulty and uncertainty of each example. The main challenges and our approaches are:

- **How can we address the uncertainties of pseudo labels for the unlabeled instances?** We generate multiple label hypotheses for each unlabeled speech instance instead of using only the 1-best prediction as a pseudo label (Section 3.1).
- **How can we train an ASR model with the multiple label hypotheses for each unlabeled instance?** We propose a novel sampling-based

loss function that incorporates multiple label hypotheses for each unlabeled instance (Section 3.2).

- **How can we adaptably address the various level of uncertainty across the unlabeled instances?** We train the ASR model beginning with easy examples, and progressing to harder ones. This provides the model with more information before encountering unlabeled instances with highly uncertain pseudo labels, enabling robust training on these instances with inaccurate pseudo labels (Section 3.3).

3.1 Multiple Hypotheses for Unlabeled Instances

MOCA utilizes an ASR model f , which is trained only with a few labeled instances, to generate pseudo labels for unlabeled instances. A straightforward approach would be to use the predicted label $\hat{l}_i$ from the trained model f as a pseudo label for an unlabeled instance x_i . However, there are inevitable uncertainties in $\hat{l}_i$, primarily due to the limited number of labeled examples, which in turn degrades the ASR performance. Another approach would be to use soft labels instead of the hard label $\hat{l}_i$ as in other deep learning area [13]. This prevents f from overfitting on the potentially incorrect predictions, therefore expected to be more robust than the previous 1-best based approach. Nevertheless, utilizing a soft labeling strategy in ASR tasks is impractical due to the intractably vast number of potential target labels. For example, the number of K -letter English words that can be formed using the alphabets is 26^K .

MOCA addresses the uncertainties associated with the pseudo labels by generating a set $Z_i = \{z_{i,1}, z_{i,2}, \dots, z_{i,N}\}$ of N label hypotheses for each unlabeled speech signal x_i . With multiple candidates for each speech instance, MOCA prevents the possibility of fully trusting the inaccurate 1-best pseudo labels. MOCA takes into account the confidence level of multiple hypotheses by evaluating the likelihoods associated with each hypothesis of a given speech instance. A higher likelihood indicates a greater probability of observing a hypothesis, making it a more reliable candidate. This can be understood in a context similar to the soft labeling strategy employed in ASR tasks with a limited set of possible target labels. This approach facilitates an efficient approximation of the computationally challenging task inherent in the naive soft labeling approach.

We define the set Z_i of label hypotheses for each x_i as the top- N candidates of labels. The top- N candidates are derived through beam search using f_θ , which is the pre-trained ASR model trained on a few labeled instances, and parameterized by θ . We define the probability $p(z_{i,j} | x_i, \theta, Z_i)$ of sampling each j -th hypothesis $z_{i,j}$ from the hypotheses set Z_i given a speech signal x_i as the normalized form of beam search scores which represent the log-likelihoods of candidates. The sampling probability of $z_{i,j}$ from Z_i is as follows:

$$p(z_{i,j} | x_i, \theta, Z_i) = \frac{e^{s_{i,j}}}{\sum_{k=1}^{|Z_i|} e^{s_{i,k}}} \quad (3)$$

where $s_{i,j}$ is the beam search score for $z_{i,j}$. Exponentiating the log-likelihood $s_{i,j} = \log p(z_{i,j} | x_i, \theta)$ transforms it into the likelihood $p(z_{i,j} | x_i, \theta)$. Thus,

$p(z_{i,j} \mid x_i, \theta, Z_i)$ in Equation (3) represents the normalized likelihood $p(z_{i,j} \mid x_i, \theta)$ within the hypotheses set Z_i .

3.2 Training ASR model with Multiple Hypotheses

Previous approaches for ASR uses the CTC loss, which is the sum of negative log likelihoods for all speech instances, to train an ASR model. For a labeled speech instance x_i with label l_i , the likelihood $p(l_i \mid x_i)$ is computed as in Equation (1). However, it is not easy to define the likelihoods for unlabeled instances since ground-truth labels are not available; we have multiple label hypotheses instead.

We define the likelihood for each unlabeled instance as the probability of observing pseudo labels sampled from the set of hypotheses Z_i . The key idea is that each element $z_{i,j}$ is selected with corresponding probabilities $p(z_{i,j} \mid x_i, \theta, Z_i)$, where θ is the parameter of the pre-trained ASR model f_θ with labeled instances. This enables MOCA to adaptably generate pseudo labels for the unlabeled instances considering their uncertainty levels.

Furthermore, sampling N-best pseudo labels from the hypotheses set enhances the robustness of the model. For instance, sampling 3 times from a set with 10 hypotheses includes diverse pseudo labels compared to that with 3 hypotheses. Considering the substantial presence of true labels within the alternative hypotheses (see Fig 1), incorporating a broader set of hypotheses as candidates for pseudo labels leads to improved pseudo label selection and enhanced robustness of the model. The distribution of certainty level among hypotheses is reflected in generated pseudo labels, and results in a more robust model.

Given a set Z_i of hypotheses for each unlabeled example x_i , the likelihood $p(S(Z_i) \mid x_i, \theta)$ of observing the sampled hypotheses $S(Z_i)$ is expressed as:

$$p(S(Z_i) \mid x_i, \theta) = \prod_{r_{i,k} \in S(Z_i)} p(r_{i,k} \mid x_i, \theta) \quad (4)$$

where $S(Z_i)$ is a list of sampled hypotheses from Z_i .

Using $p(S(Z_i) \mid x_i, \theta)$ as a likelihood for an unlabeled instance x_i , MOCA minimizes the negative log likelihood for both labeled and unlabeled instances when retraining the ASR model $f_{\theta^{\text{new}}}$ with a new parameter θ^{new} . The loss function $\mathcal{L}$ of MOCA for retraining $f_{\theta^{\text{new}}}$ is expressed as follows:

$$\mathcal{L} = - \sum_{x_i \in X_L} \log p(l_i \mid x_i, \theta^{\text{new}}) - \sum_{x_i \in X_U} \sum_{r_{i,k} \in S(Z_i)} \log p(r_{i,k} \mid x_i, \theta^{\text{new}}) \quad (5)$$

where X_L and X_U are sets of labeled and unlabeled speeches, respectively.

3.3 Curriculum Learning

If every pseudo label in the sampled hypothesis set $S(Z_i)$ for an unlabeled instance x_i is inaccurate, considering multiple pseudo labels also suffers from the uncertainty of the seed ASR model. MOCA reduces the heavy reliance on the quality of pseudo labels by implementing a curriculum learning strategy, which involves proposing a novel difficulty score for each speech instance. The main

idea is to train the ASR model with easier examples first, which are more certain ones, and train with harder and more uncertain ones later. This enables the model to easily learn the complicated decision boundary.

We propose two difficulty scores for curriculum learning. The first approach treats a speech as more challenging and uncertain when the speech is uttered rapidly or it contains a longer sentence. The first difficulty score is expressed as

$$\text{difficulty_score}(x_i) = \frac{\text{len}(l_i)}{\text{len}(x_i)} * \text{len}(l_i) \quad (6)$$

where l_i is a (pseudo) label for a speech instance x_i , $\text{len}(\cdot)$ is a function that returns the length of an input. The score is computed by multiplying the speed of a speech, which is $\text{len}(l_i)/\text{len}(x_i)$, and the length $\text{len}(l_i)$ of the uttered sentence.

The second difficulty score assesses a speech as more challenging if the spoken sentences are longer, similar to the first approach. The difference is that instances of the same length are considered easier if they have higher prediction confidence. The score is expressed as follows:

$$\text{difficulty_score}(x_i) = \text{len}(l_i) + (1 - p(l_i | x_i)). \quad (7)$$

For every instance including the unlabeled ones, MOCA computes the difficulty scores and trains the ASR model starting from examples with lower scores.

3.4 Theoretical Analysis

MOCA minimizes the loss $\mathcal{L}$ in Equation (5) to optimize the parameters θ^{new} of the ASR model $f_{\theta^{\text{new}}}$. We aim to theoretically show that the loss of MOCA is the expectation of the CTC loss in Equation (2).

Let z_i be a latent variable representing the pseudo label of x_i ; $z_i \in Z_i$ where Z_i is the set of label hypothesis for x_i . Recall that the distribution of z_i is defined with f_θ that is pre-trained with a few labeled instances (see Equation (3)). Then, the expectation of the CTC loss in terms of z_i for retraining $f_{\theta^{\text{new}}}$ on an instance x_i is expressed as follows:

$$\mathbb{E}_{z_i \sim p(z_i | x_i, \theta, Z_i)} [-\log p(z_i | x_i, \theta^{\text{new}})] = - \sum_{z_i} p(z_i | x_i, \theta, Z_i) \log p(z_i | x_i, \theta^{\text{new}}) \quad (8)$$

where $p(z_i | x_i, \theta^{\text{new}})$ is the likelihood of observing z_i using the new model $f_{\theta^{\text{new}}}$.

For each unlabeled instance x_i , MOCA selects multiple pseudo labels by sampling from the hypothesis set Z_i . Then MOCA computes the loss for x_i by summing up the negative log likelihoods of all the sampled labels (see Equation (5)). We approximate the likelihood $p(z_i | x_i, \theta, Z_i)$ in Equation (8) as the ratio of the number of times z_i is sampled to the total number sampled labels: $p(z_i | x_i, \theta, Z_i) \approx \frac{\#z_i}{|S(Z_i)|}$ where $S(Z_i)$ is a list of sampled pseudo labels from Z_i . As a result, we have the following equation:

$$\begin{aligned} \mathbb{E}_{z_i \sim p(z_i | x_i, \theta, Z_i)} [-\log p(z_i | x_i, \theta^{\text{new}})] &\approx - \sum_{z_i} \frac{\#z_i}{|S(Z_i)|} \log p(z_i | x_i, \theta^{\text{new}}) \\ &= \frac{1}{|S(Z_i)|} \sum_{r_{i,k} \in S(Z_i)} -\log p(r_{i,k} | x_i, \theta^{\text{new}}). \end{aligned} \quad (9)$$

Table 1. Summary of datasets.

Dataset	Hours	Number of speakers	Language
LJSpeech ¹	24.0	1	English
LibriSpeech-dev-clean ²	5.4	40	English
LibriSpeech-test-clean ²	5.4	40	English

¹ <https://keithito.com/LJ-Speech-Dataset/>² <http://www.openslr.org/12>

Comparing Equations (9) and (5) shows that MOCA minimizes the approximated expectation of CTC loss in terms of the unknown pseudo labels for unlabeled instances, with a balancing parameter $|S(Z_i)|$.

4 Experiments

We conduct experiments on real-world datasets with diverse settings to provide answers to the following questions.

- Q1 Transcription performance (Section 4.2).** How accurately does MOCA transcribe speech signals into texts compared to the baselines in a semi-supervised setting?
- Q2 Speed of Convergence (Section 4.3).** How does the number of hypotheses for each unlabeled instance affect the convergence speed?
- Q3 Ablation Study (Section 4.4).** Does each module of MOCA contribute to the transcription performance?

4.1 Experimental Settings.

We introduce the experimental settings including datasets, baselines, and evaluation metrics. All the experiments are performed with a single GPU machine with GTX 3080.

Dataset. We use four speech datasets summarized in Table 1. LJSpeech dataset consists of 13,100 audio clips, totaling around 24 hours of clear English speech recorded by a single female speaker. The dataset contains passages from seven non-fiction books, and a transcription is provided for each clip as a label. LibriSpeech dataset is sourced from the LibriVox project audiobooks, and categorized into dev-clean and test-clean.

Baselines. We compare MOCA with other general previous approaches for each data and difficulty score function. SUPERVISED [19] is a simple ASR model that loads pre-trained wav2vec 2.0, and finetunes it using labeled speech data. 1-BEST is an ASR model that uses the trained SUPERVISED model to generate 1-best pseudo labels for the unlabeled speech instances. SELF-TRAIN [5] is a self-training-based method for semi-supervised ASR, which generates the pseudo labels dynamically during training.

Evaluation and Settings. To evaluate the ASR performance, we use WER (word error rate) as the main evaluation metric and CER (character error rate) as an additional metric. We use wav2vec 2.0 as our ASR model, where the

Table 2. ASR performance of MOCA and baselines in terms of WER and CER. Bold and underlined numbers indicate the best and the second-best performance, respectively. Note that MOCA with various settings outperforms the competitors.

Model	# of sampled hypotheses K	Difficulty score	LJSpeech		dev-clean		test-clean	
			WER	CER	WER	CER	WER	CER
SUPERVISED	N/A	N/A	10.08	2.23	21.60	6.75	19.68	6.19
1-BEST	N/A	N/A	8.21	1.74	16.22	5.28	15.20	4.30
SELF-TRAIN	N/A	N/A	8.99	2.01	16.16	5.34	17.16	5.07
MOCA	3	confidence	<u>6.85</u>	<u>1.44</u>	15.23	4.80	15.11	<u>4.19</u>
	3	speed	6.88	1.48	15.32	4.94	15.01	4.22
	5	confidence	6.76	1.43	15.69	4.87	<u>15.05</u>	4.10
	5	speed	6.98	1.51	<u>15.21</u>	4.72	15.32	4.48
	10	confidence	<u>6.85</u>	1.47	15.86	4.72	15.24	4.23
	10	speed	8.87	2.14	14.83	<u>4.76</u>	15.75	4.63

initial parameters are set following [17]. We split LJSpeech and LibriSpeech datasets into training-labeled, training-unlabeled, and test sets with ratio 0.1 : 0.4 : 0.5 and 0.5 : 0.4 : 0.1, respectively, considering the small number of speech instances in LibriSpeech. The training-labeled set comprises speech instances with transcriptions, whereas the training-unlabeled set consists of those without transcriptions. We set the number of candidates in the hypotheses pool to 10, and vary the number of sampled pseudo labels in $\{3, 5, 10\}$.

4.2 Transcription Performance (Q1)

We compare the transcription performance of MOCA and the baselines in Table 2. Note that MOCA shows significant improvement in terms of both WER and CER than the 1-BEST model. Specifically, MOCA with the number $K = 5$ of sampled pseudo labels presents higher transcription accuracy than MOCA with $K = 3$. The result indicates that considering more numbers of pseudo labels from the pool of hypotheses improves the prediction performance while too large K confuses model with many possible answers, degrading the performance.

4.3 Speed of Convergence (Q2)

To study the effect of the number K of pseudo labels for each unlabeled instance in MOCA, we show the change of transcription performance as the epoch proceeds in Fig 3. MOCA- K -conf and MOCA- K -speed represent MOCA with confidence-based and speed-based difficulty scores for curriculum learning, respectively, while having K sampled pseudo labels in common. Fig 3 shows that all cases of MOCA converge in earlier epochs than 1-BEST, which uses the 1-best hypothesis as a pseudo label for each unlabeled instance. This indicates that considering multiple hypotheses enhances the speed of convergence and transcription performance. Also note that a larger K in MOCA results in faster convergence. This is due to the significant presence of alternative hypotheses containing true labels; incorporating more hypotheses as candidates leads to improved selection of pseudo labels and enhances the robustness of the model.

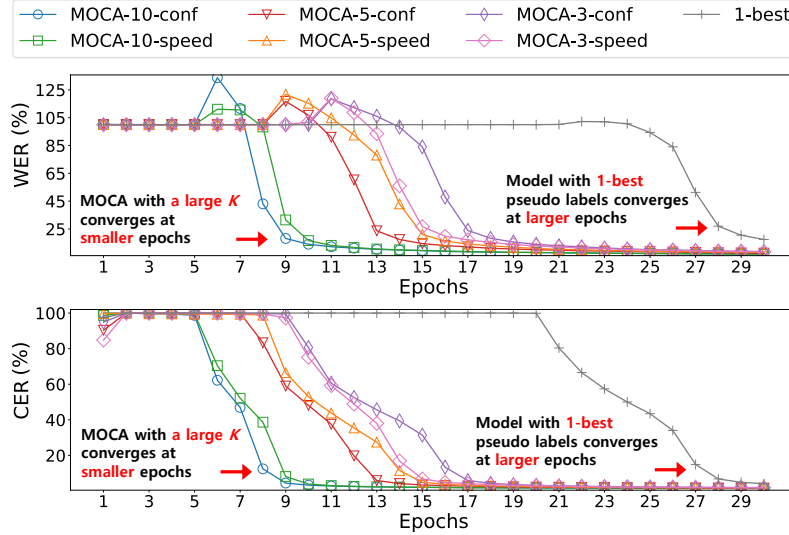


Fig. 3. Transcription performance of MOCA as the epoch proceeds in LJSpeech dataset. MOCA always converges at smaller epochs than the 1-best model, which uses a single pseudo label for each unlabeled one. Also note that more number of pseudo labels for each unlabeled instance results in quicker convergence.

4.4 Ablation Study (Q3)

We study the contribution of each module in MOCA for transcription performance in Table 3. MOCA-1-best and MOCA-uniform-sampling refer to MOCA without considering multiple pseudo labels, and MOCA that uniformly samples pseudo labels from the hypotheses set, respectively. MOCA-w/o-curriculum and MOCA-inverse-curriculum indicate MOCA without curriculum learning, and MOCA with the reversed learning order of instances, respectively. We set the number K of sampled hypotheses as 10, and use the speed-based difficulty score for curriculum learning through the ablation study.

Table 3 shows that each module of MOCA effectively contributes to the transcription performance. MOCA consistently outperforms MOCA-1-best and MOCA-uniform-sampling in terms of error rates, highlighting the significance of sampling pseudo labels based on sampling probabilities from the hypotheses set. MOCA demonstrates better performance compared to MOCA-w.o.curriculum and MOCA-w.inverse-curriculum, with MOCA-w.inverse-curriculum showing the worst performance. This indicates that the proposed curriculum score effectively improves the transcription performance.

5 Conclusions

We propose MOCA, an accurate method for semi-supervised ASR. Unlike previous approaches that overlook the uncertainty associated with pseudo labels for unlabeled instances, MOCA considers the uncertainty by incorporating multiple

Table 3. Ablation study. Bold numbers indicate the best performance. Note that each module contributes to the transcription performance.

Model	dev-clean		test-clean	
	WER	CER	WER	CER
MOCA-1-best	15.76	4.93	16.83	5.19
MOCA-uniform-sampling	16.20	4.84	23.19	6.60
MOCA-w/o-curriculum	15.29	4.98	16.08	4.59
MOCA-inverse-curriculum	18.72	5.81	21.67	6.36
MOCA	14.83	4.76	15.75	4.63

hypotheses through sampling from a pool of probable hypotheses. Furthermore, MOCA relieves the problem of the dependence on the quality of the multiple pseudo labels by implementing a curriculum learning strategy with our novel difficulty scores. Experiments on real-world datasets show that MOCA outperforms other baselines in terms of transcription accuracy by addressing uncertainties in pseudo labels.

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